

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: CELL DIVISION (SUB-STRAND 1.4)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How does the body produce the enormous number of new cells needed for growth? Think back to the baby photographs and the time-lapse of a child growing over 14 years: a newborn of about 3 kg became a teenager of 50 kg or more. Use what you observed and learned in Lessons 1 to 10 to write your own Final Explanation. Answer each section in the blank space under the prompt, in full sentences. Use evidence from our investigations (diagrams, clay models, onion root tips, animations, case studies) and give reasons for your claims. You may add labelled sketches. Use correct scientific words such as chromosome, chromatid, daughter cell, mitosis, meiosis, zygote and gamete.

1. Claim: Where do the new cells come from?

A newborn of about 3 kg grows into a teenager of 50 kg or more, yet we only eat food such as ugali, sukuma wiki and beans. (a) Write a claim that answers the driving question. (b) Explain why eating food alone cannot account for the billions of new cells. (c) Compare this with your first ideas in Lesson 1 and say how your thinking has changed.

Write your answer here:

2. Evidence: Mitosis and growth

Use evidence from the mitosis diagram, the clay model and the onion (kitunguu) root tip squashes. (a) Describe what you saw happen to the chromosomes as one cell became two, giving the stages of mitosis in the order they happen. (b) In the onion root tip, most cells were in interphase and a smaller number were dividing. What does this tell you about growing tissues? (c) Explain how a single fertilised cell could become billions of cells, using the pattern 1, 2, 4, 8, 16...

Write your answer here:

3. Reasoning: Why the new cells work properly

(a) What does a chromosome carry, and what is the structure of a chromosome before a cell divides? (b) Explain why DNA must be copied before a cell divides. (c) Explain how this makes sure each new body cell has the right instructions to work properly. (d) Name two body processes besides growth that need mitosis.

Write your answer here:

4. Meiosis and reproduction

Every baby begins as one fertilised cell (zygote) formed when a sperm meets an egg. (a) If a sperm and an egg each had 46 chromosomes, how many would the zygote have, and what problem would that cause over generations? (b) Describe what meiosis does to the chromosome number and how many cells it makes. (c) Explain how meiosis and mitosis work together so that a child can be produced and then grow. (d) Give one difference between the cells made by mitosis and meiosis.

Write your answer here:

5. Other forms of cell division and their uses

(a) Describe how budding in yeast and binary fission in bacteria differ. (b) A single bacterium dividing every 20 minutes gives how many cells after 2 hours? Show how you worked it out, and say how this doubling is similar to what happens in mitosis in a growing child. (c) Explain how tissue culture of bananas or tea in Kenya uses the same idea as growth in a child. (d) Finish by writing a short Final Explanation (3 to 4 sentences) answering the driving question.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that new cells come from division of existing cells, and clearly shows that food provides only energy and materials, not cells directly.	States that cell division makes new cells, but the role of food is unclear or only briefly mentioned.	Claim is missing, vague, or suggests food turns directly into cells or that cells only grow bigger.
Use of evidence from the investigations	Uses several specific observations (mitosis diagram, clay model, onion root tips, the 1, 2, 4, 8, 16 pattern) accurately to support the claims, and correctly treats the 64-bacteria figure as a calculation that shows the same doubling pattern rather than as an observation.	Uses one or two observations from the lessons, mostly accurate but general.	Gives little or no evidence, or evidence is inaccurate, not from the lessons, or calculations are presented as observations.
Scientific reasoning about chromosomes, mitosis and meiosis	Correctly explains DNA copying (chromatids identical after copying), the stages of mitosis in order, identical daughter cells, and how meiosis halves the chromosome number (23) so the zygote has 46; links both to the growing child.	Explains mitosis and meiosis mostly correctly but misses a link (for example the order of stages, or why DNA is copied or why the number is halved).	Confuses mitosis and meiosis or does not explain how new cells get the right instructions.
Breadth: growth, repair,	Connects cell division to growth, repair,	Covers growth and at least one other	Mentions only growth, or applications are

reproduction and applications	replacement, sexual and asexual reproduction (budding, binary fission) and tissue culture with accurate Kenyan examples.	function or application accurately.	missing or incorrect.
Communication and use of scientific terms	Writes organised, complete sentences with correct terms (chromosome, chromatid, daughter cell, zygote, gamete) and a clear final explanation.	Mostly clear writing with some correct terms; final explanation is complete but may be loosely organised.	Writing is hard to follow, key terms are missing or misused, and the final explanation is incomplete.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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